Leveraging LLMs for Efficiency in Image Generation

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Why InstanceDiffusion

Introducing InstanceDiffusion to enhancing text-to-image diffusion models by providing precise instance-level control.

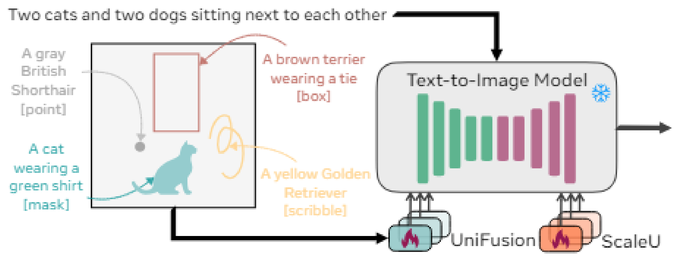


Fig 1: InstanceDiffusion

Enabling precise instance-level control through the three learnable blocks:

1. UniFusion: enables instance-level conditions for text-to-image models.
2. ScaleU : improves image fidelity.
3. Multi-instance Sampler: improves generations for multiple instances.

Reproduction of Experiments

We reproduced results with a subset of 500 images from the evaluation set, representing 10% of the original dataset. This subset was selected due to the computational expense associated with generating new images via InstanceDiffusion. Examples of these images are displayed below.

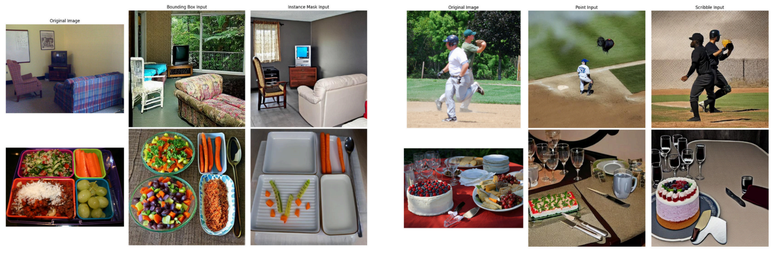


Fig 2: Random samples generated by bounding boxes, instance masks, points and scribbles

Method	AP_{box}	AP_{box}^{50}	AR_{box}
InstanceDiffusion	38.8	55.4	52.9
Our Reproduction	49.9	66.8	68.6
Difference	+11.1	+11.4	+15.7

Table 1. Evaluating different location formats when generating images of reproduction experiments using Bounding Boxes as input.

Method	AP_{mask}	AP_{mask}^{50}	AR_{mask}
InstanceDiffusion	27.1	50.0	38.1
Our Reproduction	40.8	63.5	56.0
Difference	+13.7	+13.5	+17.9

Table 2. Evaluating different location formats when generating images of reproduction experiments using Instance Masks as input.

Methods	Color		Texture	
	A_{cc}^{color}	$CLIP_{local}$	$A_{cc}^{texture}$	$CLIP_{local}$
InstanceDiffusion	54.4	0.250	26.8	0.225
Our Reproduction	53.3	0.248	26.9	0.226
Difference	-1.1	-0.002	+0.1	+0.001

Table 4. Attribute binding reproduction results for color and texture.

Leveraging LLM's for Modular Efficiency in InstanceDiffusion.

In the UniFusion block, a GPT-4 LLM was incorporated to automate the generation of image descriptions and bounding boxes. Utilizing the LLM, we address the complexity of interpreting and executing prompts with multiple objects and spatial relationships.

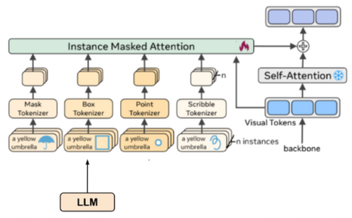


Fig 3: Incorporation of the LLM in the UniFusion block

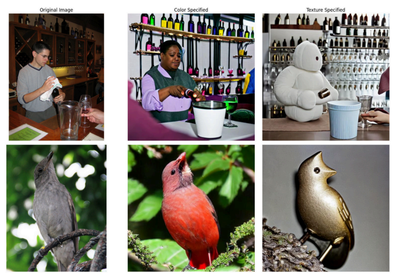


Fig 4: Images of color and texture

Evaluating of LLM Submodule

Evaluated the LLM submodule, using two approaches:
1. Multiple evaluators assessed the realism of the generated images.
2. Deployed a robust Vision Language Model (VLM) called CogVLM to determine if the generated images fulfilled the input descriptions.

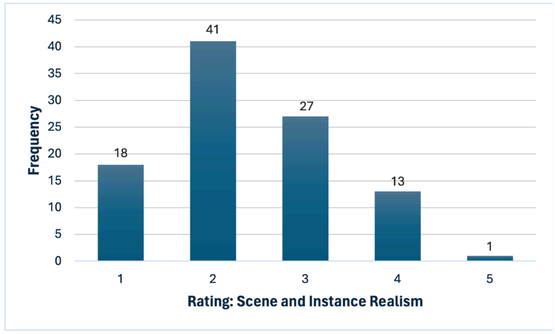


Fig 5: Distribution of Ratings from all evaluators



Fig 6: Side-by-side visualizations of bounding box predictions and ground truths for object detection. (GPT-4 and CogVLM)

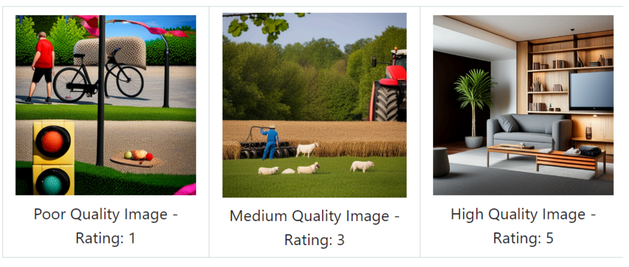


Fig 7: Example of manual evaluators assesments